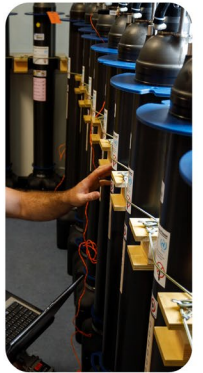
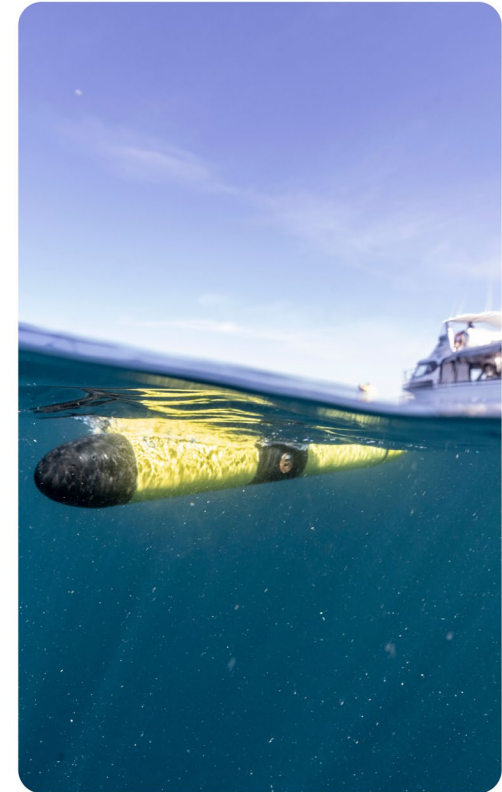
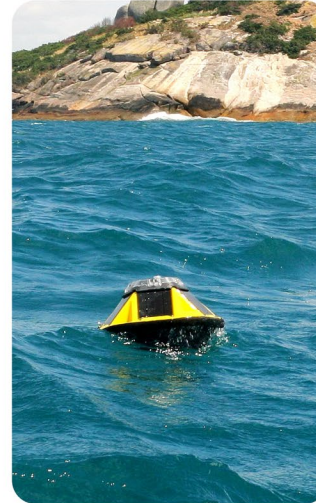


AMSA 2025 IMOS Workshop

Working with GIS

Benjamin Stepin

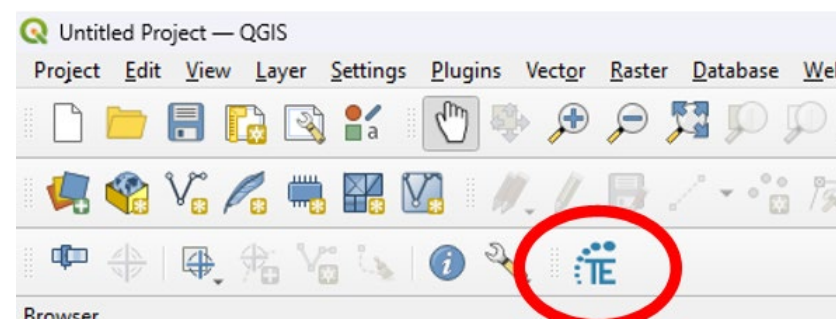
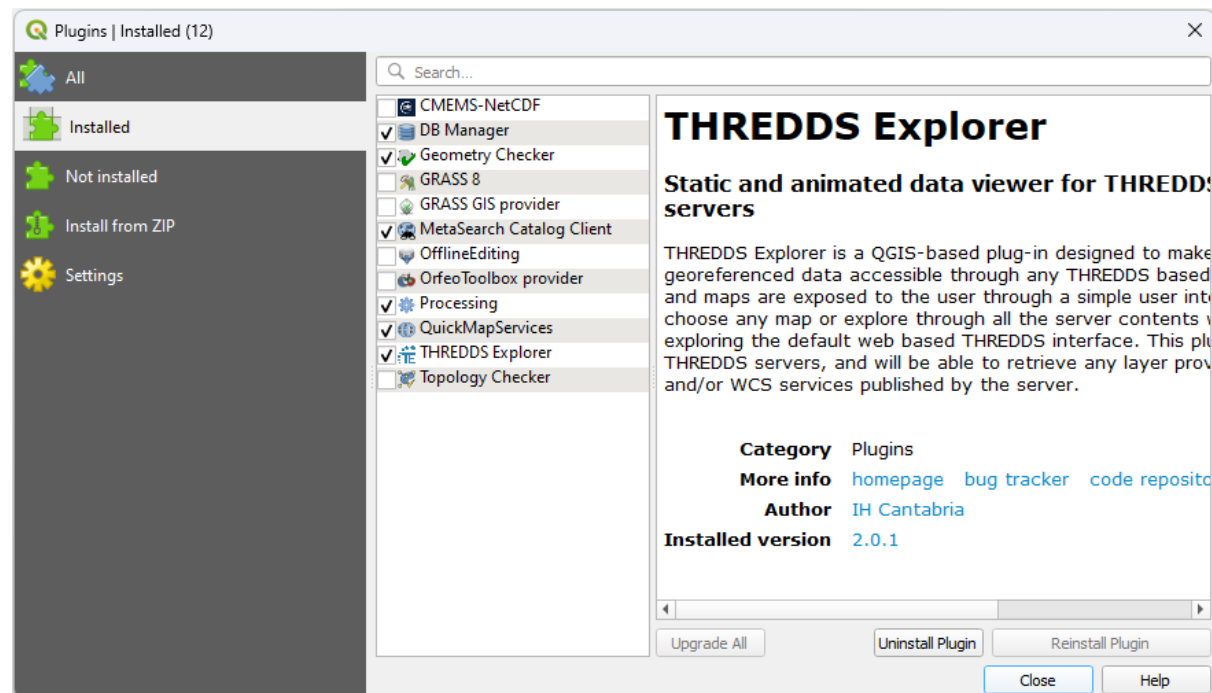
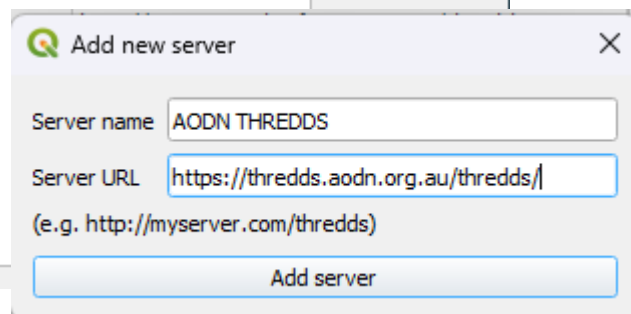
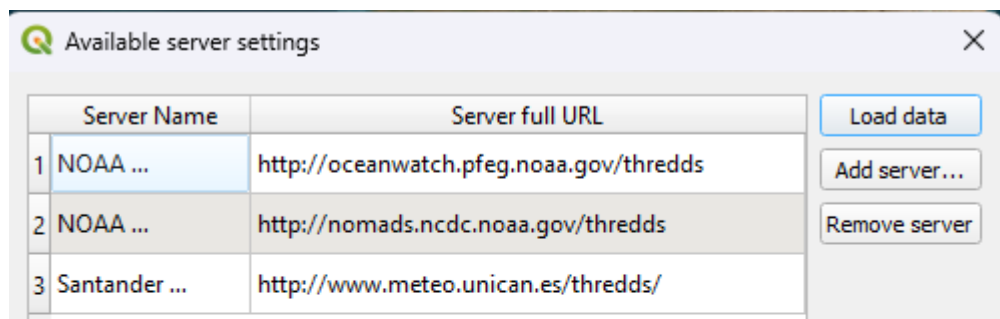


IMOS acknowledges the Traditional Custodians and Elders of the land and sea on which we work and observe, and recognise them as Australia's first marine scientists and carers of Sea Country. We pay our respects to Aboriginal and Torres Strait Islander peoples past and present.

Discover Data

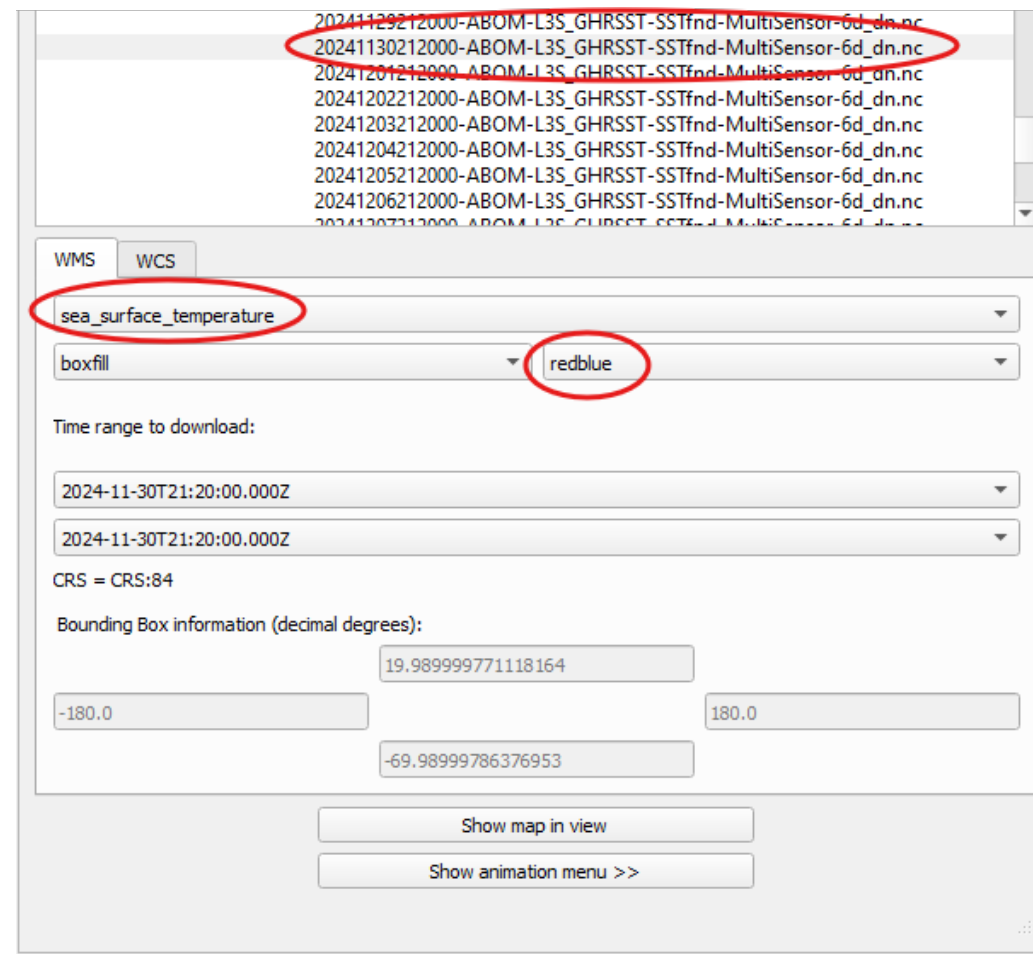
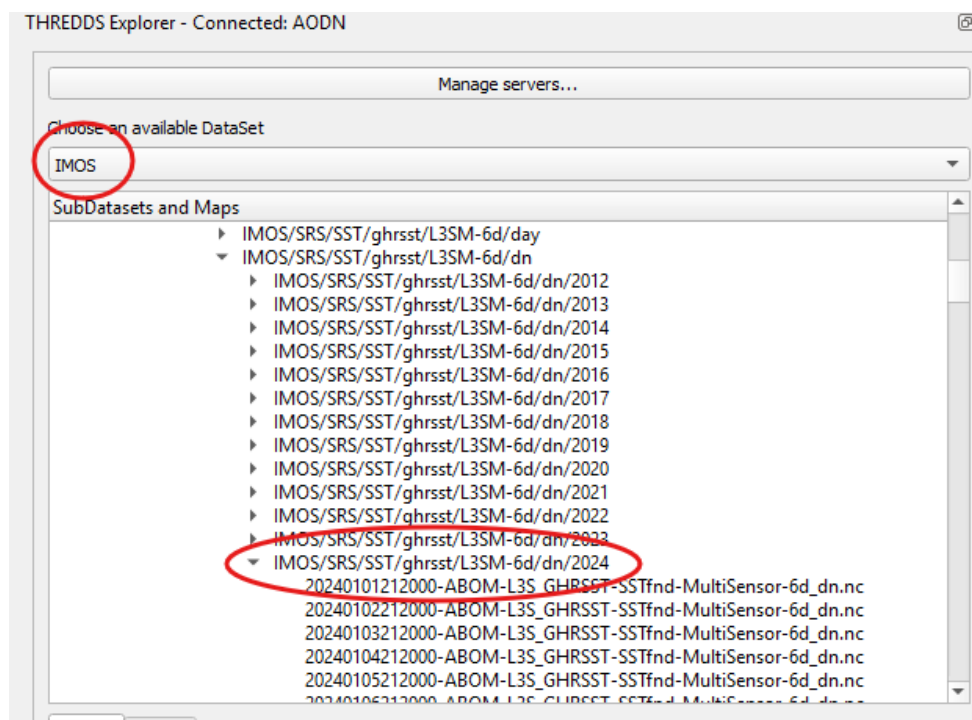
Easiest way to discover data that you might want to download and process is using the “THREDDS Explorer” plugin in QGIS.

Instructions can be found in the GIT repository of the plugin.



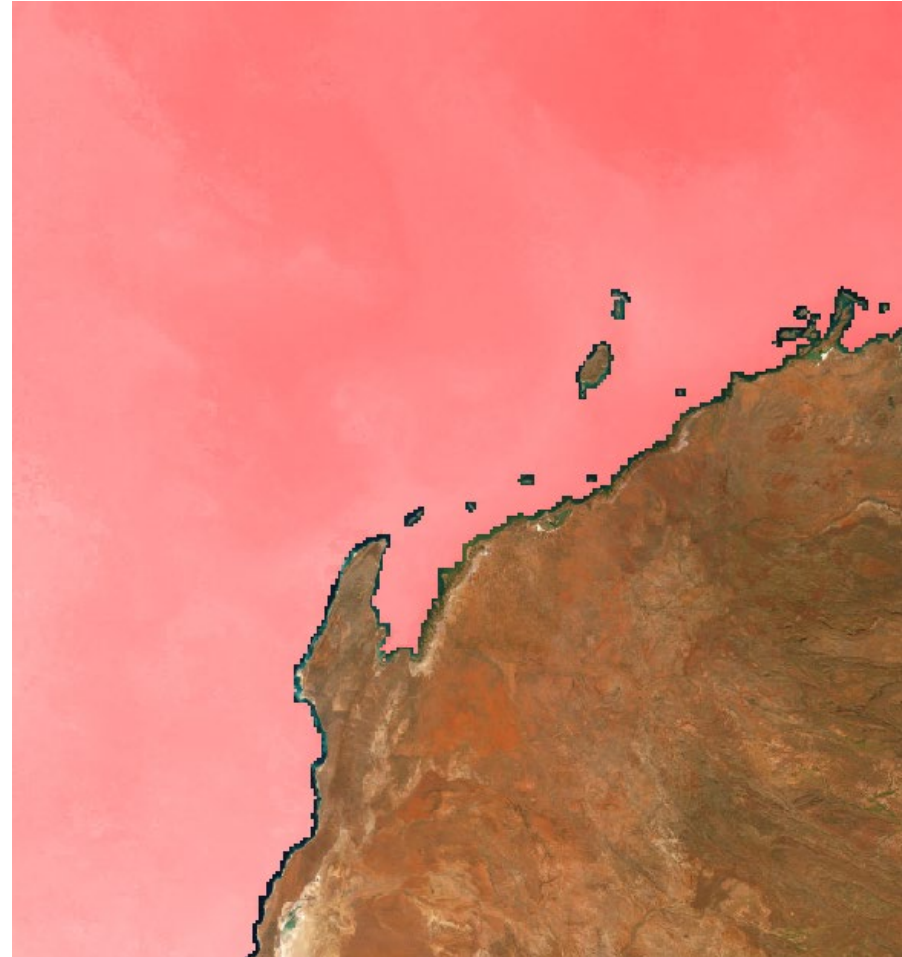
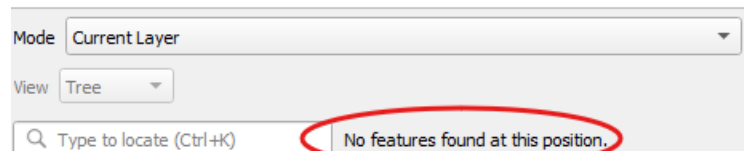
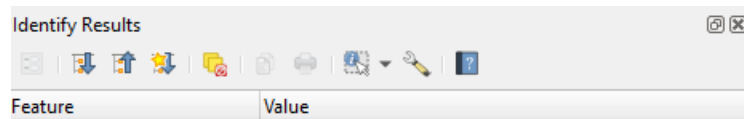
Data Discovery

Let's assume we would like to observe on the heatwave towards the end of 2024. We can navigate to the SRS product in TE. Navigate to IMOS/SRS/SST/ghrsst/L3M-6d/dn/2024 And find a date you would like to observe.



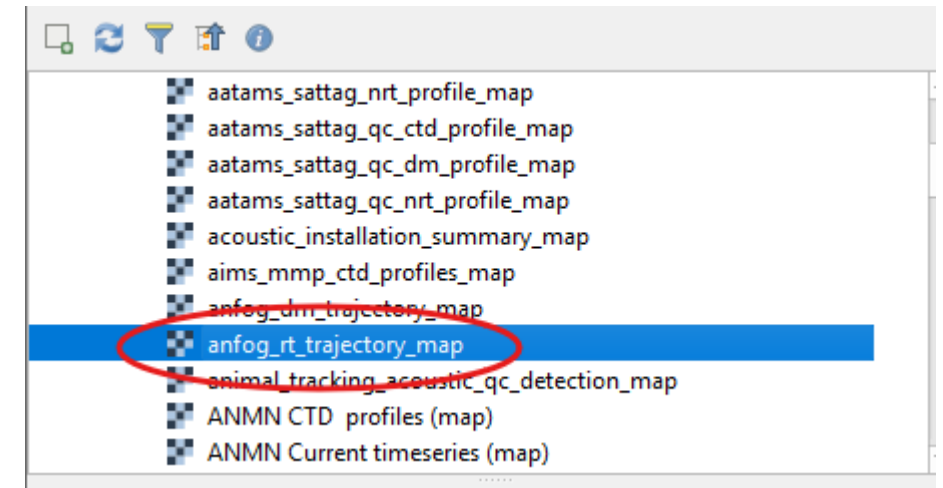
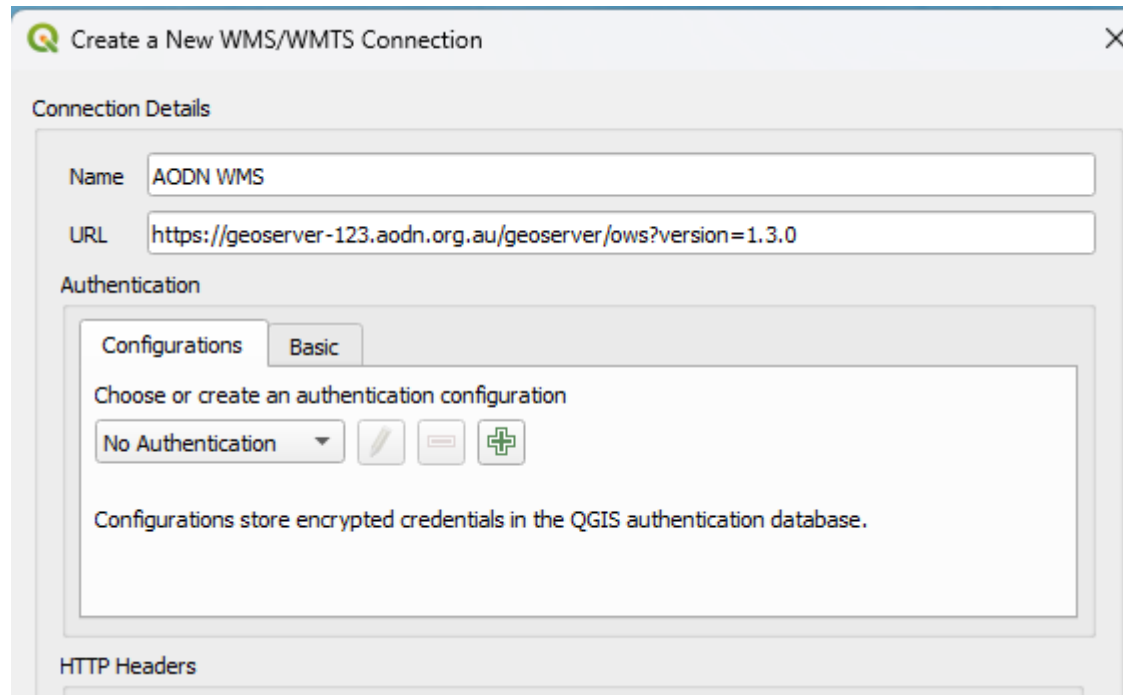
Visualisation Only

- THREDDS Explorer does not import the actual data behind the file.
- Allows to explore available data quickly.
- Has some good features.



Other tools

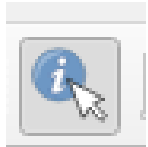
- Other tools include direct connection to WMS and WFS services
- Connect to WMS with the URL:
<https://geoserver-123.aodn.org.au/geoserver/ows?version=1.3.0>



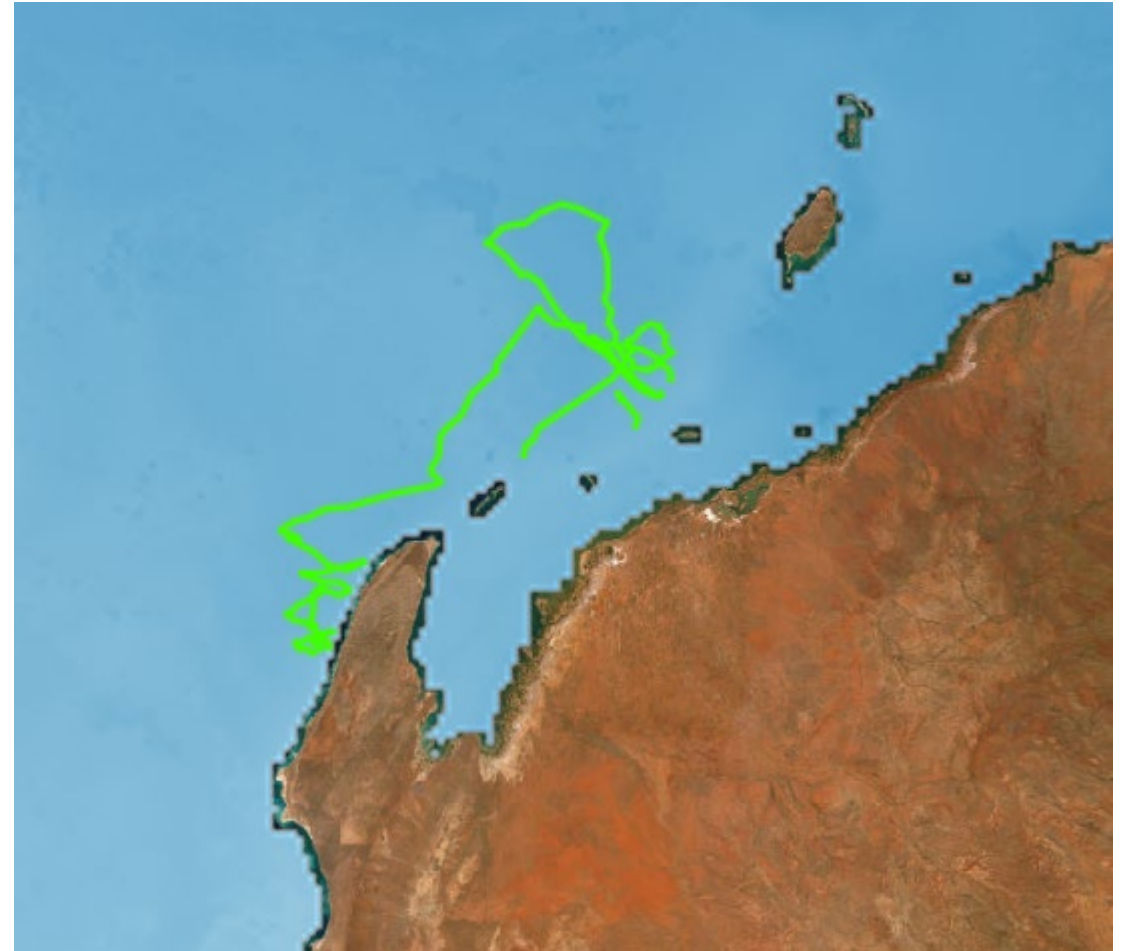
AODN WMS

The Web Mapping Service (WMS) provides track data

Allows to identify time the tracks were made



Feature	Value
▼ anfog_rt_trajectory_map	0
Format	HTML
▼ imos:anfog_rt_trajecto...	
Glider Deployment : Onslow20241119	
Slocum glider : SL994	
Deployment status: This deployment is in progress	
Time range : Nov 19, 2024, 1:25:03 AM - Dec 11, 2024, 12:06:14 AM	
View charts of this deployment: Charts	



Download SST data

We can head to the portal to download the data we will be using today as an example.
First the SST in the data range 19/11/24 – 12/12/24 Download the NetCDF file.

Step 1: Select a Data Collection

New Search

Parameter

- ☒ Physical-Water
 - ☒ Temperature
 - Skin temperature of the water body (38)
 - Temperature of the water body (3)

Organisation

- ☒ Australian Universities (1)
- ☒ Commonwealth Departments and Agencies (41)
- ☒ Integrated Marine Observing System (IMOS) (40)

Platform

- ☒ Satellite
 - geostationary orbiting satellite (4)
 - orbiting satellite (38)

Date (UTC)

Geographic Boundary

Keyword

Type keyword, press enter

IMOS - Satellite Remote Sensing - SST - L3S - Multi Sensor - 6 day - day and night time - Australia

Temperature
2012 - 2025
Integrated Marine Observing System (IMOS) - Australian Bureau of Meteorology
Satellite

IMOS - Satellite Remote Sensing - SST - L3S - Multi Sensor - 3 day - night time - Australia

Temperature
2012 - 2025
Integrated Marine Observing System (IMOS) - Australian Bureau of Meteorology
Satellite

IMOS - Satellite Remote Sensing - SST - L3S - Multi Sensor - 3 day - day time - Australia

Temperature
2012 - 2025
Integrated Marine Observing System (IMOS) - Australian Bureau of Meteorology
Satellite

IMOS - Zooplankton Abundance and Biomass Index (CPR)

Chlorophyll - Ocean Biota - Temperature
2007 - 2023
Integrated Marine Observing System (IMOS) - CSIRO Oceans and Atmosphere - Hobart
Satellite - Vessel

Step 2: Create a Subset

Spatial

N enter

Bounding Box

W enter E enter

Reset

S enter

IMOS - Satellite Remote Sensing - SST - L3S - Multi Sensor - 6 day - day and night time - Australia

Subset Info Layer

Temporal

From 2024-11-19 21:20 UTC

To 2024-12-12 21:20 UTC

Move Time on Map

Reset 'Dates'

Displaying: 2024-12-12 21:20:00:000 UTC

☐ Point timeseries

Latitude enter Longitude enter

Clear Subsets

Download Glider data

Next download the glider track 19/11/24 – 12/12/24 and the bounding box around WA. Download the CSV.

Step 1: Select a Data Collection

New Search

Parameter

Physical-Water

Temperature

Temperature of the water body (4)

Organisation

Australian Universities (3)

Commonwealth Departments and Agencies (2)

Integrated Marine Observing System (IMOS) (2)

State and Territory Government Departments and Agencies (1)

Platform

Glider

sub-surface gliders (4)

Date (UTC)

Geographic Boundary

Keyword

Type keyword, press enter

IMOS - Ocean Gliders - delayed mode glider deployments

Chlorophyll - Circulation - Optical properties - Oxygen - Salinity - Temperature - Water pressure

2008 - 2025

Integrated Marine Observing System (IMOS) - Department of Jobs, Tourism, Science and Innovation (DJTSI), Western Australian Government - Oceans Graduate School

IMOS - Ocean Gliders - Real-time glider deployments

Chlorophyll - Circulation - Optical properties - Oxygen - Salinity - Temperature - Water pressure

2010 - 2025

Integrated Marine Observing System (IMOS) - Oceans Graduate School (OGS), The University of Western Australia (UWA)

Defence Science and Technology Organisation (DSTO) - delayed mode glider deployments

Circulation - Salinity - Temperature

2011 - 2014

Defence Science and Technology Organisation (DSTO), Department of Defence (DoD), Australian Government - Maritime Division, Defence Science and Technology Group (DST Group)

Slocum glider measurements of the Australian North West Shelf in April 2012

Chlorophyll - Optical properties - Other Pigment - Salinity - Temperature - Turbidity

2012 - 2012

UWA Oceans Institute (OI), The University of Western Australia (UWA) - Australian Institute of Marine Science (AIMS)

Step 2: Create a Subset

Spatial

N -19.34

Bounding Box

W 112.72 E 117.03

Reset

S -23.29

IMOS - Ocean Gliders - Real-time glider deployments

Subset

Info

Layer

Temporal

From 2024-11-19

To 2024-12-12

Reset

Others

Deployment Name

All

Platform Type

All

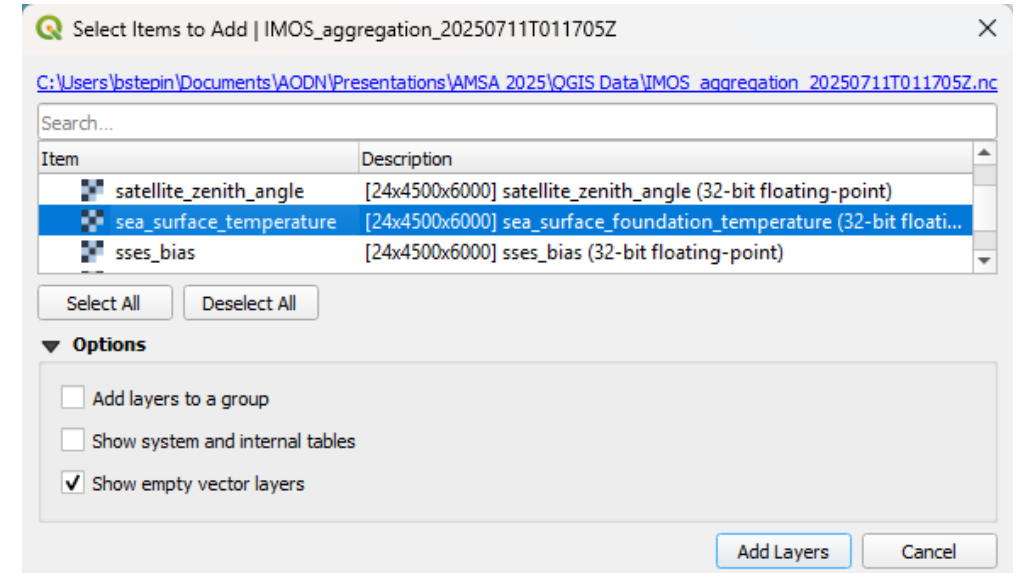
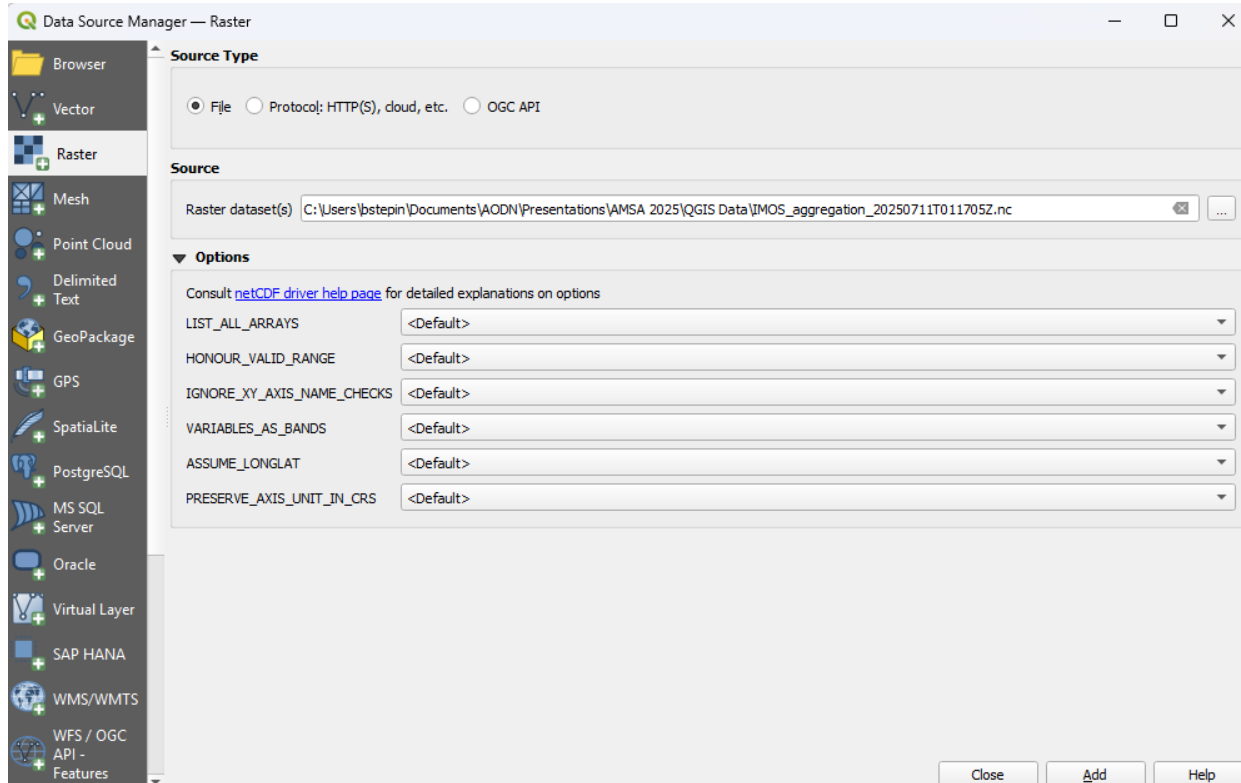
Status

All

Clear Subsets

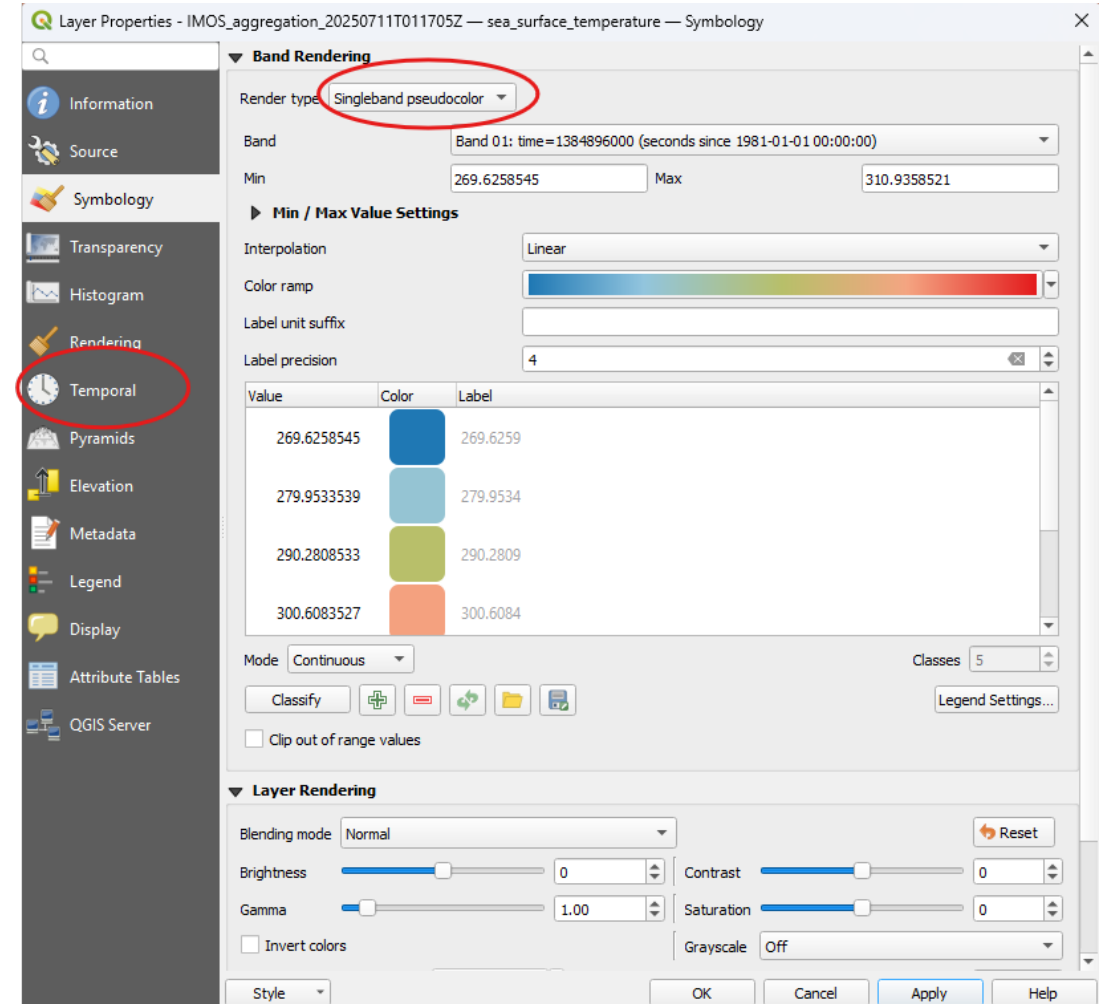
 IMOS Integrated Marine Observing System

Load in the NetCDF file to the project



Refining the import

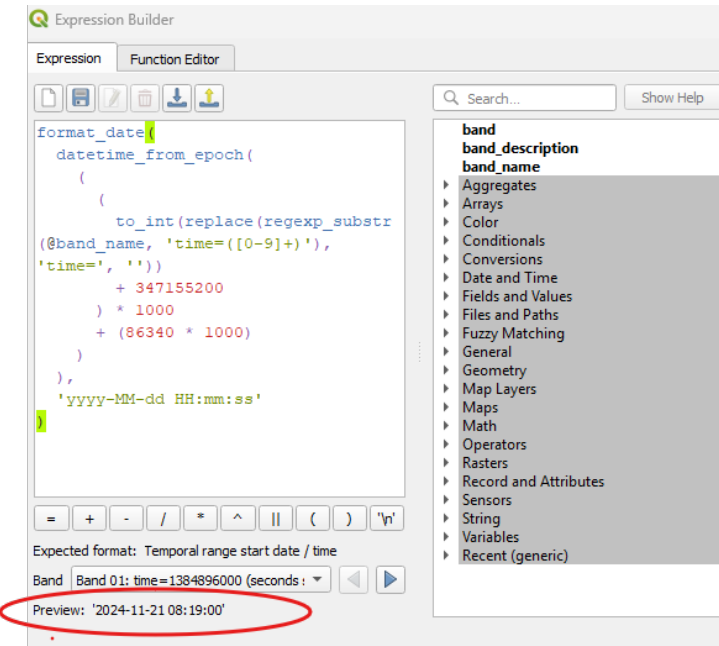
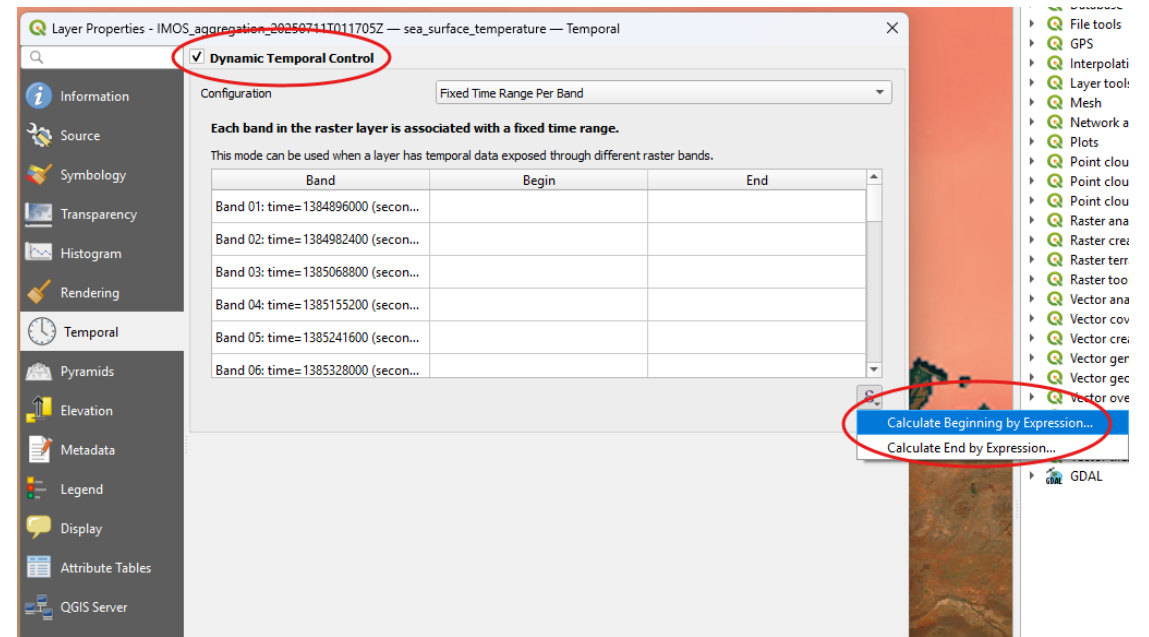
- Imports multiband NetCDF
- Each band represents a day
- Set symbology to Single Band Pseudocolor
- Adjust colour to your liking
- NOTE. Apply the change by pressing “OK” not just “Apply”



Heading

- Head to temporal tab
- Head to the Expression calculator and enter the formula across.
- This Extracts the date from the band title, and formats it to the correct date/time format (seconds since 1/1/1981 0:00)
- Datetim_from_epoch is the seconds since 1/1/1970 0:00 we add the 347155200 seconds to make the date 1981.

```
format_date(
  datetime_from_epoch(
    (
      to_int(replace(regex_substr(@band_name, 'time=([0-9]+)'), 'time=', ''))
      + 347155200
    ) * 1000
  ),
  'yyyy-MM-dd HH:mm:ss'
)
```



Add Glider data

Add “Delimited Text”

- Set the number of Header lines to 41, as the rest is embedded metadata.
- Ensure “Trim Fields”, “First record has headers” and “Detect field types” are enabled
- Ensure X and Y field are correctly assigned

Data Source Manager | Delimited Text

File name: C:\Users\jstepin\Documents\AODN\Ppresentations\AMSA 2025\QGIS Data\IMOS_-_Ocean_Gliders_-_Real-time_glider_deployments 2024.csv

Layer name: IMOS_-_Ocean_Gliders_-_Real-time_glider_deployments 2024

Encoding: UTF-8

File Format

- ☒ CSV (comma separated values)
- ☐ Regular expression delimiter
- ☐ Custom delimiters

Record and Fields Options

Number of header lines to discard: 41

☒ First record has field names

☒ Detect field types

☒ Trim fields

☐ Discard empty fields

Custom boolean literals

True: False:

Geometry Definition

☒ Point coordinates

X field: LONGITUDE

Y field: LATITUDE

☐ Well known text (WKT)

☐ No geometry (attribute only table)

☐ DMS coordinates

Geometry CRS: EPSG:4326 - WGS 84

Layer Settings

☒ Use spatial index

☐ Use subset index

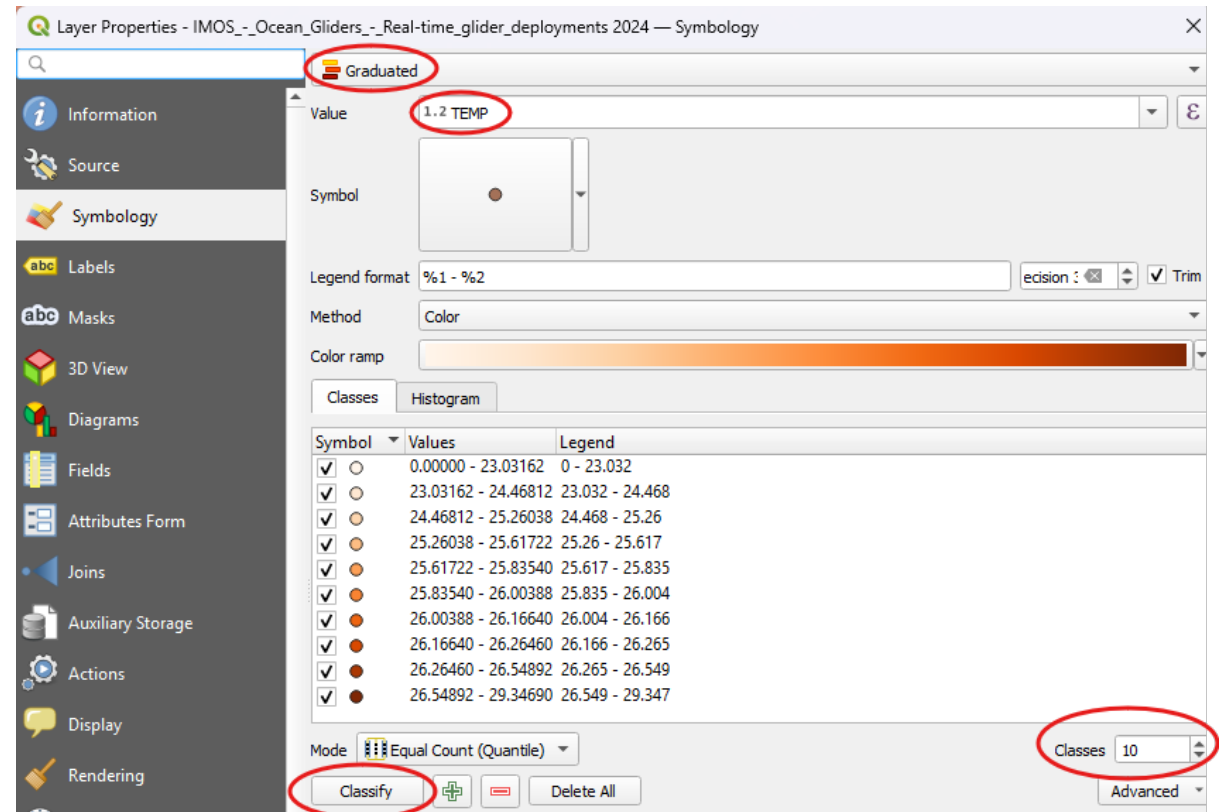
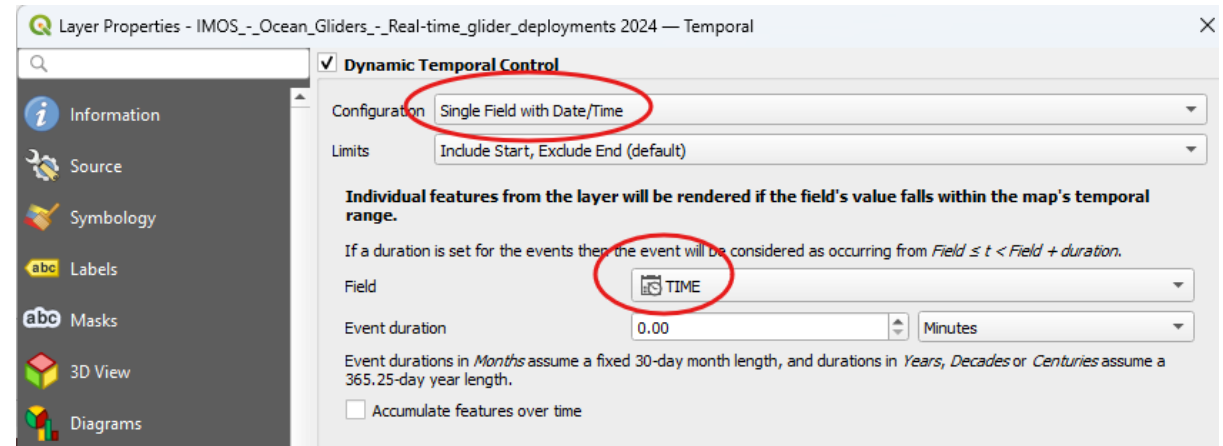
☐ Watch file

Sample Data

	FID	file_id	deployment_name	platform_code	platform_type
	abc Text (string)	123 Integer (32 bit)	abc Text (string)	abc Text (string)	abc Text (string)
1	anfogl_trajectory_data.fid--462e4c35_197f7005023_5537	52511	Onslow20241119	SL994	slocum glider
2	anfogl_trajectory_data.fid--462e4c35_197f7005023_5538	52511	Onslow20241119	SL994	slocum glider
3	anfogl_trajectory_data.fid--462e4c35_197f7005023_5539	52511	Onslow20241119	SL994	slocum glider
4	anfogl_trajectory_data.fid--462e4c35_197f7005023_553a	52511	Onslow20241119	SL994	slocum glider
5	anfogl_trajectory_data.fid--462e4c35_197f7005023_553b	52511	Onslow20241119	SL994	slocum glider
6	anfogl_trajectory_data.fid--462e4c35_197f7005023_553c	52511	Onslow20241119	SL994	slocum glider
7	anfogl_trajectory_data.fid--462e4c35_197f7005023_553d	52511	Onslow20241119	SL994	slocum glider

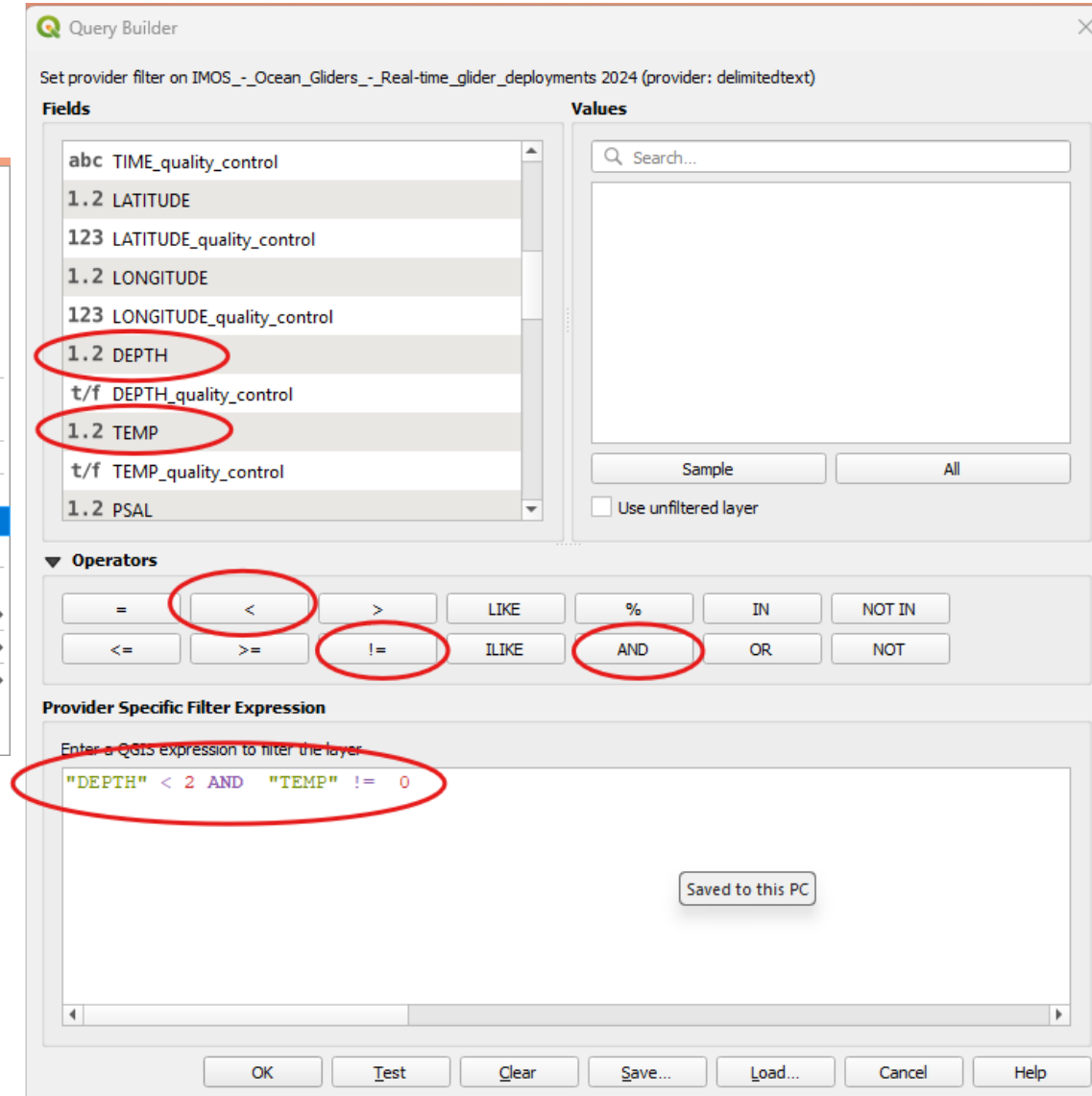
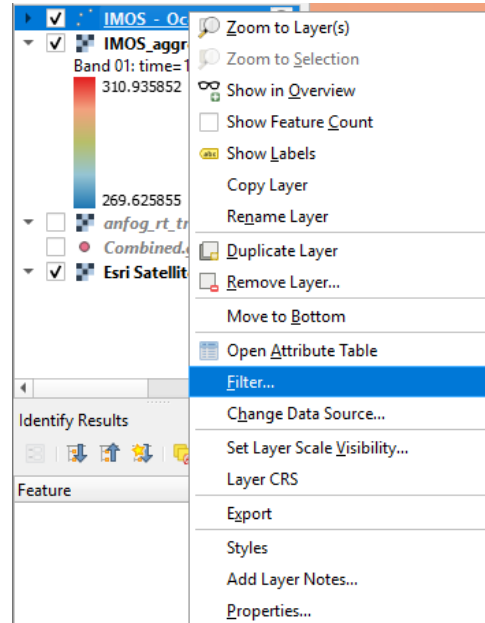
Adjust Data

- Once added, go to layer properties and head to the Temporal tab
- Set single field with Date/Time and select “TIME” as the Field
- Go to the Symbology Tab and set to “Graduated”, Set value to “TEMP”, choose a colour ramp and then Hit “Classify”. (Using 10 in example)

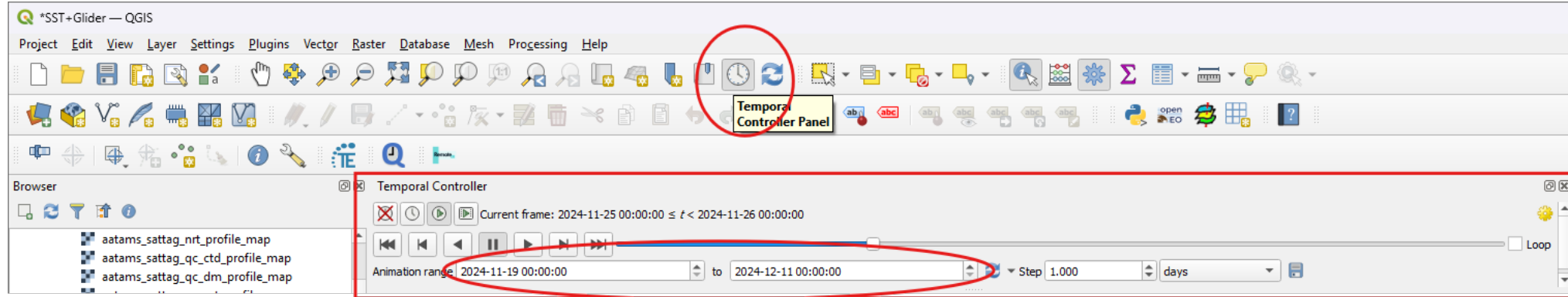


Filter to Surface Data

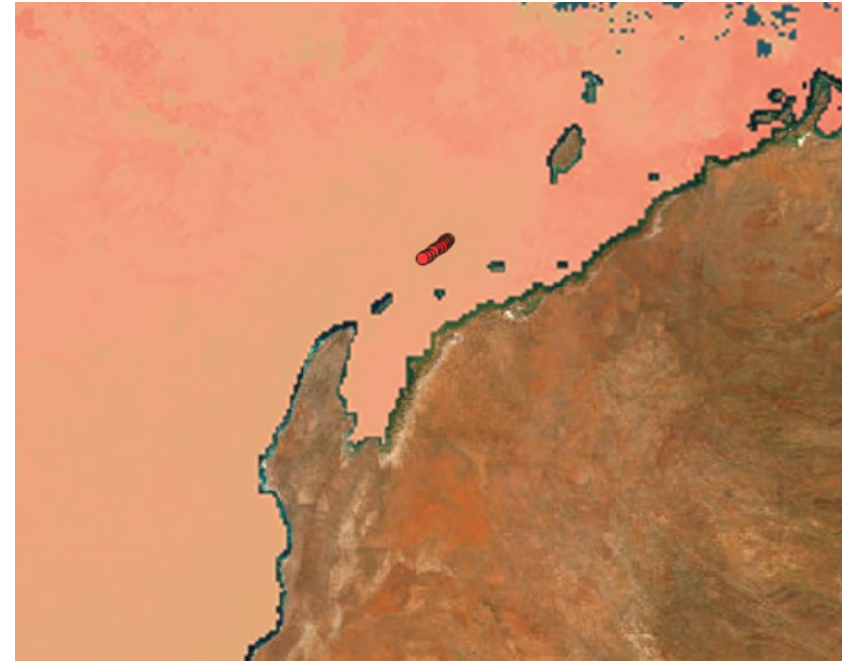
- In order to compare only surface data we can filter to only observe the measurements the glider has undertaken at the surface.
- In this case we are assuming surface to be anything less than 2m depth
- We're also not interested in values where the TEMP is set to 0 so use and "AND" operator.



Activate the Temporal Controller

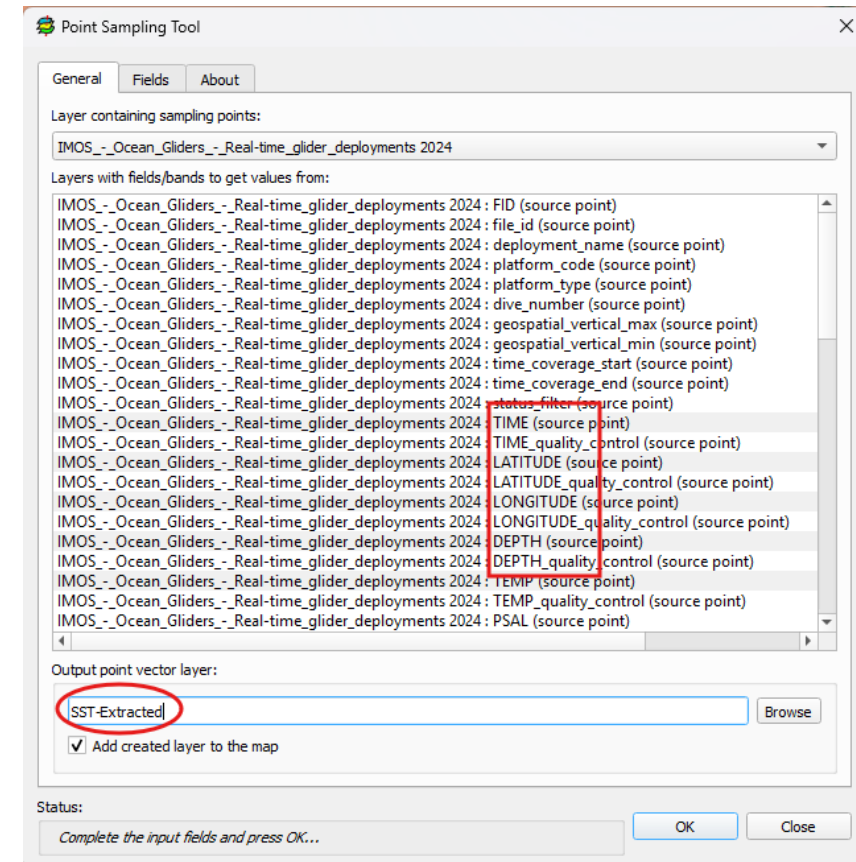
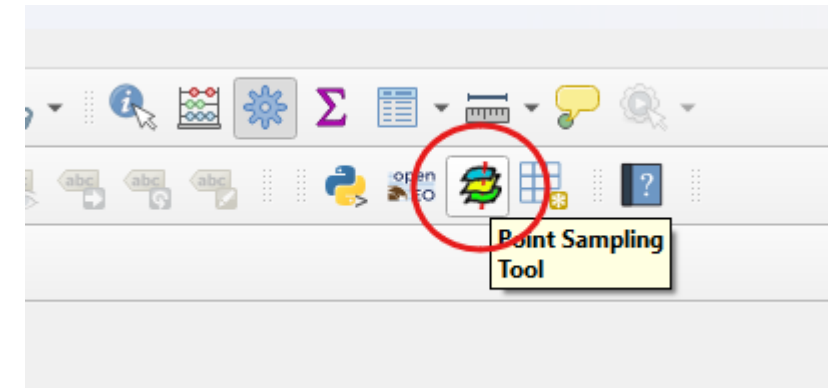


- Open the Temporal Controller, set to the required date range and the increment to 1day.
- Now you can cycle through the glider days as well as the SST on that day and observe the changes.



Extract the SST Values

- If you've installed the "Point Sampling Tool" navigate to it to extract the points.
- From the Gliders select the values we would like to keep, noting the "Filtered" will not be extracted.
- In this example we are keeping "TIME" Lat/Lon, "DEPTH" and "TEMP"
- From the SST we are using the middle static band12. You can extract all the bands and then create an expression later in a similar fashion to the temporal expression we used earlier to extract the SST band corresponding to the date.



Correct from Kelvin to Celsius

- Open the newly created layer “Attribute table”
- Open the Field Calculator, and create a new field
- Select the SST field and subtract 273.15 to convert to C.
- Name the column appropriately.
- Now this can be exported to CSV

SST-Extracted.gpkg — Features Total: 342, Filtered: 342, Selected: 0

123 fid = 123

Open field calculator (Ctrl+I)

	fid	TIME	LATITUDE	LONGITUDE	DEPTH	TEMP	711T011705Z — s
1	1	19/11/2024 02:0...	-21.329	114.9102	0.1193	26.29	300.1249
2	2	19/11/2024 02:3...	-21.3284	114.9118	0.3776	26.3419	300.1249

SST-Extracted.gpkg — Field Calculator

☐ Only update selected feature(s)

☒ Create a new field

☐ Create virtual field

Output field name: SST (C)

Output field type: Real (Decimal number (real))

Output field length: 0 Precision: 3

Expression: "IMOS_aggregation_20250711T011705Z — sea_surface_temperature_12" - 273.15

Function Editor

row_number

Aggregates

Arrays

Color

Conditionals

Conversions

Date and Time

Fields and Values

feature

geometry

id

NULL

fid

TIME

1.2 LATITUDE

1.2 LONGITUDE

1.2 DEPTH

1.2 TEMP

1.2 IMOS_aggreg...

Files and Paths

OK Cancel Apply Help

SST-Extracted.gpkg — Features Total: 342, Filtered: 342, Selected: 0

123 fid = 123

	fid	TIME	LATITUDE	LONGITUDE	DEPTH	TEMP	711T011705Z — se	SST (C)
1	1	19/11/2024 02:0...	-21.329	114.9102	0.1193	26.29	300.1249	26.9749...
2	2	19/11/2024 02:3...	-21.3284	114.9118	0.3776	26.3419	300.1249	26.9749...
3	3	19/11/2024 03:3...	-21.3268	114.9111	-0.0795	26.4592	300.1249	26.9749...
4	4	19/11/2024 04:0...	-21.3262	114.9094	-0.0596	26.5345	300.1249	26.9749...
5	5	19/11/2024 04:3...	-21.3257	114.9076	-0.0596	26.6194	300.1249	26.9749...
6	6	19/11/2024 05:0...	-21.3252	114.9056	-0.0795	26.7484	300.1249	26.9749...
7	7	19/11/2024 06:0...	-21.3285	114.9042	-0.0596	26.7216	300.1249	26.9749...
8	8	19/11/2024 07:0...	-21.332	114.9026	-0.0298	26.7368	300.1249	26.9749...

Save Vector Layer as...

Format: Comma Separated Value [CSV]

File name: bstepin\Documents\AODN\Ppresentations\AMSA 2025\QGIS Data\SST-Glider-Compare.csv

Layer name:

CRS: EPSG:4326 - WGS 84

Encoding: UTF-8

☐ Save only selected features

☒ Select fields to export and their export options...

☒ Persist layer metadata

Geometry

Geometry type: Automatic

☐ Force multi-type

☐ Include z-dimension

☐ Extent (current: none)

Layer Options

CREATE_CSVT: NO

GEOMETRY: <Default>

LINEFORMAT: <Default>

SEPARATOR: COMMA

STRING_QUOTING: IF_AMBIGUOUS

WRITE_BOM: NO

☒ Add saved file to map

OK Cancel Help



Australia's Integrated Marine Observing System is enabled by the National Collaborative Research Infrastructure Strategy (NCRIS). It is operated by a consortium of institutions as an unincorporated joint venture, with the University of Tasmania as Lead Agent.

PRINCIPAL PARTICIPANTS



SIMS is a partnership involving four universities

ASSOCIATE PARTICIPANTS



IMOS thanks the many other organisations who partner with IMOS, providing co-investment, funding and operational support, including investment from the Tasmanian and Western Australian Governments.

